

Temporal Associative Memory

Online sequence learning

The “temporal” associative memory component learns higher-order sequences and next-token prediction on the fly.

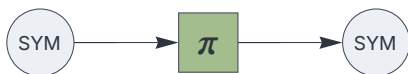
Requirements

```
<< "Circuits.m"
```

Circuit

```
{
  "$schema": "https://creatingintelligence.org/schemas/circuit-v1.json",
  "hyperparameters": {"default": [3000, 10]},
  "dataflow": [
    {"component": "input", "plugin": "codec", "send": [11]},
    {"component": "temporal", "send": [21], "receive": [11], "capacity": 8, "decimate": 0.8},
    {"component": "output", "plugin": "codec", "receive": [21]}
  ]
}
```

```
circ = Circuit[json];
c = circ["function"];
circ["schematics"][ImageSize→220]
```



Dataset

Generate a 2000 character random string.

```
chars = CharacterRange["A", "Z"]
```

```
{A, B, C, D, E, F, G, H, I, J, K, L, M, N, O, P, Q, R, S, T, U, V, W, X, Y, Z}
```

```
dataset = RandomChoice[ chars, 2000];
```

```
StringJoin[dataset]
```

```
XTXVNWYKFIDCHGGUDAMZYMQXIOPUAKYOCZLGJQJKAPXMIUJLSHLZOMGTWWTGNMGGVUFLDWJT  
VBTRUBYJCOHZSLOCVKWBCWQHTCTBDSUEQKCHOAVIWOKVIVURGBXJKDIDQQBYANZWAHSIG  
GTUFXNKXSZFOEGGCIZRBTEODUAFGPYULCSQGSJTKUGHTMPRBVUFTPXETKIGDALSWNTLH  
HTTQUIESZXYOTGFTRLVCFMDUJUBHPGKDVVYZHHTHITHGQNBWCIUDEUHLQAIDPMOXQOSI  
LVLJONBVRABDRNTZYSOZPKULFHFUHKYPADRYUPINNKNHJRIFCKUQKILCDHBDVQEYCIBFC  
JLBPTZAEMLLFAAJIEJZZXNIFKQBFWLLKYOZOASHJRHKXYETWEPYQOAPXRBCBLNDXAJON  
AASEXZFHEAVMYMOAVIRNCZEBLGQVJVZJPXLCMDBTUMOCUGWWXGGYUESQLVGXDTGNUEQLE  
FDTSPKXKRCTLYIJKRUEYPLVEUZURJLVBWTRNQSSUVYGNXINYOVXAWUABXRQUNVETWUB  
BLJKTUUSHZKKLFZPDJORWYLYTBFQORRDUKPGYQCJMKXEUFGBMBHVJQITIXEWRFWERKZXSN  
IBFEWQNLJRNGAWQBAEEKIMHMCQGEZQPGINYYKMUMZAMUHFAUGEEIWJIAZFWWQEQEJSKUS  
CQWCPOHOLOQPTJUQHOFJSMXUGQCMRUEMMEPHUNIBHKWSRCNKQVZJAUNOPXYWUIRXWQRZN  
GLYRWTHYJRUQINECIFOFWEBLLMLPEDXEBXEIUAFLTLGFKVXPKLCIBUZQBBQVFISCBLEVM  
JJHSGZWVKXZSSBEMMHSWZQULILOOZOVKSFQFHSXPHARNGXFMXAOXRYYGUHZPEHSLKOG  
SWZVUHTQEDQZZNGGLCPGBXDVUVSEIANKDPCXHVQAJOWZUENKUNLDMIUEVNDDESICLCYQ  
IJXBNILONJUGADLWGPZYCFPHQMDIIDTPAPOIRAARZOSLIYKTUZFEUIDHPLDFSHGRTJGT  
LTOFQBDSTNZNDOFBFTRAWTMJXTYISCSBSLMPWUYTJAOVWVODQOOOJRTFEOPVPSZGJMMSQM  
XZAUCOBAHPWVANUJJITZTTENZQNWYTPYGQOWOHGRCYVHVKBKSEKOBMJOJNFUWWHNRTTM  
HAFQQXNTJUTVRBKYQRSXDVUZKZABWFKHQPIXOYIWVRVRHLNFWAHHYZJIDRJFSSYFCHLJN  
ONDJLWKNXVFZWREKKUOSJXXDASKLIBJEOZGTNRLNYQSDNPZAPMJHZTLOKKADAHELDLWFI  
FXIZXZDKPRISWEIUMRBGFJXXHXHARMNTJENMOPHGLUGJTFVCTSZMFAXYHRCTEDQIBUTCU  
WPSYNYWROMPHGCMABPHVFEJXXVKIHBMFHTIMBVEHDHIUTQBVMKGYMLCEPRNNYJUOKSOS  
QHFTDQGAKEIPJKTAAHIZHIHTVFRZPFSVHWLFETMDVIURIPFLOTQUWVXHJZVAPRDUNTGXZA  
PBPPREBTRSGFPWLRJDLPADBKHDFBFYFTUDOQQLKDQQQGBQSSQOBRBOMFSTBBQXEMKJRN  
UMDDCSUHNEIDFHRXZGQRAOBQGRDXDOVJDBFROBEJUABTLFXQEJSSXQGCVWFKATHQYZ  
RZNWXWGZYSFSZUQWHCFCNRMPPKVM SHNKNOMPBTYVMTIDOUKLXLGEIBUZHJLJPVIQGTIB  
SMTROIVYKFLBARMPOABFAUFJXGXGFFYTXVCFDYUWVCHTJNYPBAEVDIUMBUSZLGTJQYHWK  
VJXMYSGWOZOIAUUPVPLPDANYFJNVGLWYRLDTOMMFDMEAVXUIKOWANRNUCWQMDFYJDQMKN  
NWEUVIACYZIFWCBFYFOEWAQCNTXLOPOZXIGELLACSNVBAIKONCTDDBHICCFRFFKMNQOMR  
JXAFCDQSBZFCKSUUMCVGYYIWDQONZFCQSNHUDVEWMTIBYOYSFGWSACJGQLUDUUQO
```

Prediction score

```
Do[
  predictions = c /@ dataset;
  Print["  iteration = ", i,
    ", prediction score = ", N[Mean@Boole@MapThread[SameQ,
      {dataset, RotateRight[predictions]}]]],
    {i, 1, 20}];
```

```
iteration = 1, prediction score = 0.038
iteration = 2, prediction score = 0.9685
iteration = 3, prediction score = 0.9945
iteration = 4, prediction score = 0.9965
iteration = 5, prediction score = 0.9985
iteration = 6, prediction score = 0.9995
iteration = 7, prediction score = 0.998
iteration = 8, prediction score = 0.9985
iteration = 9, prediction score = 0.998
iteration = 10, prediction score = 0.997
iteration = 11, prediction score = 1.
iteration = 12, prediction score = 0.9985
iteration = 13, prediction score = 0.999
iteration = 14, prediction score = 0.9995
iteration = 15, prediction score = 0.999
iteration = 16, prediction score = 0.999
iteration = 17, prediction score = 1.
iteration = 18, prediction score = 1.
iteration = 19, prediction score = 0.999
iteration = 20, prediction score = 0.9995
```